This is the Sedimentologika LaTeX template example and build test

Abstract

This is a test abstract used for evaluating the build of an article using the *Sedimentologika* LaTeX template using LuaLaTeX. It contains no meaningful scientific content, but contains useful guidelines for writing your article using the template.

If your author and affiliation lists and CRediT roles are long, the abstract may run over the first page in 'preprint' mode. If this is the case, please attempt to shorten your abstract; however, if it is not reasonably possible please contact support@sedimentologika.org. Use \documentclass[preprint]{sedimentologika} to test the length of your abstract.

Plain language summary

This file tests the *Sedimentologika* article template build using LuaLaTeX. A plain language summary is mandatory.

Translations

We encourage authors to provide a translation of your abstract and plain language summary into a language of your choice. If you wish to provide multiple translations, please contact the editorial team for guidance

If your translation language requires special fonts (e.g. Arabic), instructions are available in the template file and the README.md file. Should you need further help with this, please contact the editorial team support@sedimentologika.org.

1 Introduction

As of 2026, *Sedimentologika* has moved to LaTeX to typeset accepted publications in the journal. The primary motivations for this switch are to:

- use an open-source, free software for typesetting,
- provide authors with a streamlined pre-print to publication workflow,
- lower the time investment required by our volunteer typesetting team, and,
- ensure all papers published in *Sedimentologika* have a unified design.

While the primary uses of the template are for typesetting by the *Sedimentologika* team and authors preparing their article for submission to *Sedimentologika* it also provides you with an easy way to generate a pre-print version that you can post on a pre-print server (e.g. EarthArXiv). Simply use \documentclass[preprint]{sedimentologika} in the preamble above.

Like the policy of Seismica, we do not require authors to use this template at initial submission. **However**, minimum formatting requirements apply and upon acceptance authors using LaTeX should submit their final version using the *Sedimentologika* LaTeX template. Not using the template may incur delays in copy-editing, proofing, and publication, For authors using Microsoft Word or LibreOffice etc, there may be greater delays incurred depending on the complexity of the document.

2 Minimum formatting and layout requirement

While *Sedimentologika* strongly encourages the use of our templates for articles submitted to the journal, you may submit your article without using the templates if the following conditions are met:

- Your article is written in a single dialect of English
- Your manuscript is a single PDF, DOC, DOCX, or ODT file containing all figures and in-text tables.
- Your data is made accessible via a repository such as Zenodo.
- You have added to your manuscript the following statements, after the conclusions and before the references
- Acknowledgements
- Data and code availability
- Conflict of interest statement
- Artificial intelligence use statement
- Funding declaration

3 Language

All submitted documents must be in English. Authors are required to use a single variety of English consistently throughout the manuscript, such as Australian, British, Canadian, Commonwealth, or US English.

While *Sedimentologika* requires manuscripts to be written in English, we encourage authors to provide an abstract and plain language summary translation in their preferred language of choice.

Submitted documents should be thoroughly proof-read prior to submission to ensure they are free from grammatical and spelling errors.

Abbreviations that are not standard to the field are not to be used in the abstract and should be introduced in the corpus of the text. The use of standard abbreviations such as "Fm." or "Mbr" respectively, for "Formation" or "Member" should be reserved for figures and tables, while full-length words are used in the corpus of the manuscript.

All units must be from the International System of Units or derivative thereof (i.e. metres rather than feet or kPa rather than PSI). Customary/non-SI units may be included in parentheses.

Utilise the commands and options available in siunitx for any quantity or number.

- \ang[options]{angle}
- \num[options]{number}
- \unit[options]{unit}
- \qty[options]{number}{unit}
- \numlist[options]{numbers}
- \numproduct[options]{numbers}
- \numrange[options]{numbers}{number2}
- \qtylist[options]{numbers}{unit}
- \qtyproduct[options]{numbers}{unit}
- \qtyrange[options]{number1}{number2}{unit}

3.1 Inclusive Language

Sedimentologika strongly encourages authors and co-authors to use inclusive and non-discriminatory language in their manuscripts. The American Psychological Association (APA) has produced a guide you may find useful for this purpose, which you can obtain from apa.org.

Scientific research is always a team effort. Therefore, we encourage using "we" about the authors as much as using passive voice or third person.

Sedimentologika encourages authors to use both traditional place names and their English equivalents, where appropriate, to acknowledge the cultural significance and diversity of the regions being described, e.g. ”the Kaikōura Canyon is located east of Te Wāhipounamu (the South Island of Aotearoa/New Zealand)”.

When applicable, acknowledgement of traditional custodians and their lands should be added.

3.2 Politically Sensitive Geographic Names

As Earth Scientists, we often use geospatial data. In some cases the geographic name of a location may be disputed, or the border of a territory has not been honoured or is disputed.

For cases of geographic name disputes (e.g. Gulf of Mexico/Gulf of America), authors should first determine if an arbitration decision has been made, for example under the UNCLOS, and use that name. If no such decision has been made, authors should use the name which is most recognised internationally, refer to appropriate United Nations maps (e.g. UN Map of the World), or provide both names. The intent is to ensure the name used is as neutral as possible.

3.3 Geologic Time

For geologic time, please use nomenclature of the International Chronostratigraphic Chart, the latest version can be found at text.

4 Figures and tables

This section outlines the general guidelines for figures and tables. Please refer to *Sedimentologika*'s figure and table guidelines for further detail.

The final document will have a two-column page layout. You can test the appearance of your tables and figures in this layout by using the `preprint` option for the `sedimentologika` document class.

4.1 Dimensions

It is best practice to design and test your figures and tables at their intended final size. Figures and tables should preferably be designed to fit within either a singe column (e.g. Figure 1) or across two columns (e.g. Figure 2) in portrait page mode. However, landscape figures and tables are allowable, but only use these if absolutely required for legibility. Table 1 summarises the maximum dimensions allowable for figures and tables that fit within a single page.

Intended Width	Maximum Width	Maximum Height
one column	86 mm	230 mm
two column	180 mm	230 mm
landscape	257 mm	160 mm

Table 1 – Summary of standard dimensions for figures and tables.

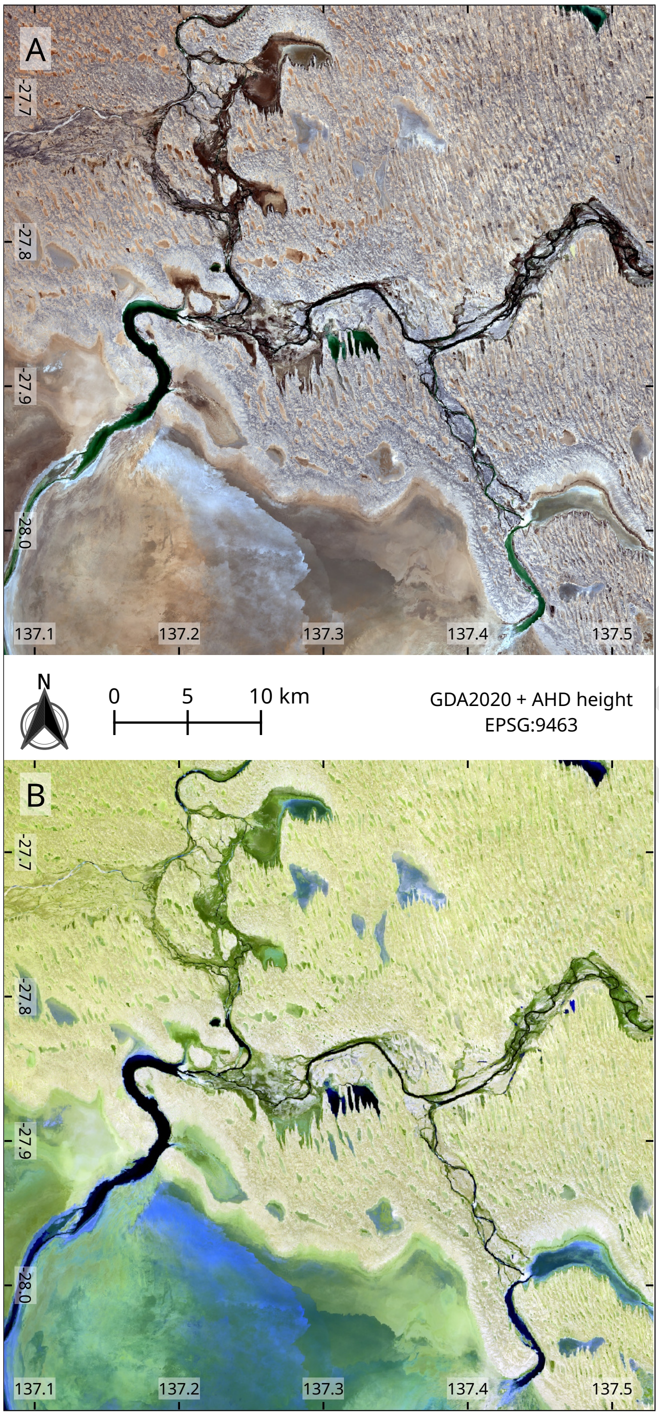

Figure 1 – Sentinel 2 L2A imagery (2025-10-18) of north-eastern Kati Thanda-Lake Eyre at the Warburton River discharge. **(A)** True colour image mapping bands 4, 3, and 2 to R, G, and B channels. **(B)** False colour image mapping bands 12, 11, and 2 to R, G, and B channels. Contains modified Copernicus Sentinel data [2025].

If a figure is being placed at the end of the document rather than near its insertion point define the figure optional position argument using a `!` or use `\clearpage` directly after the figure environment.

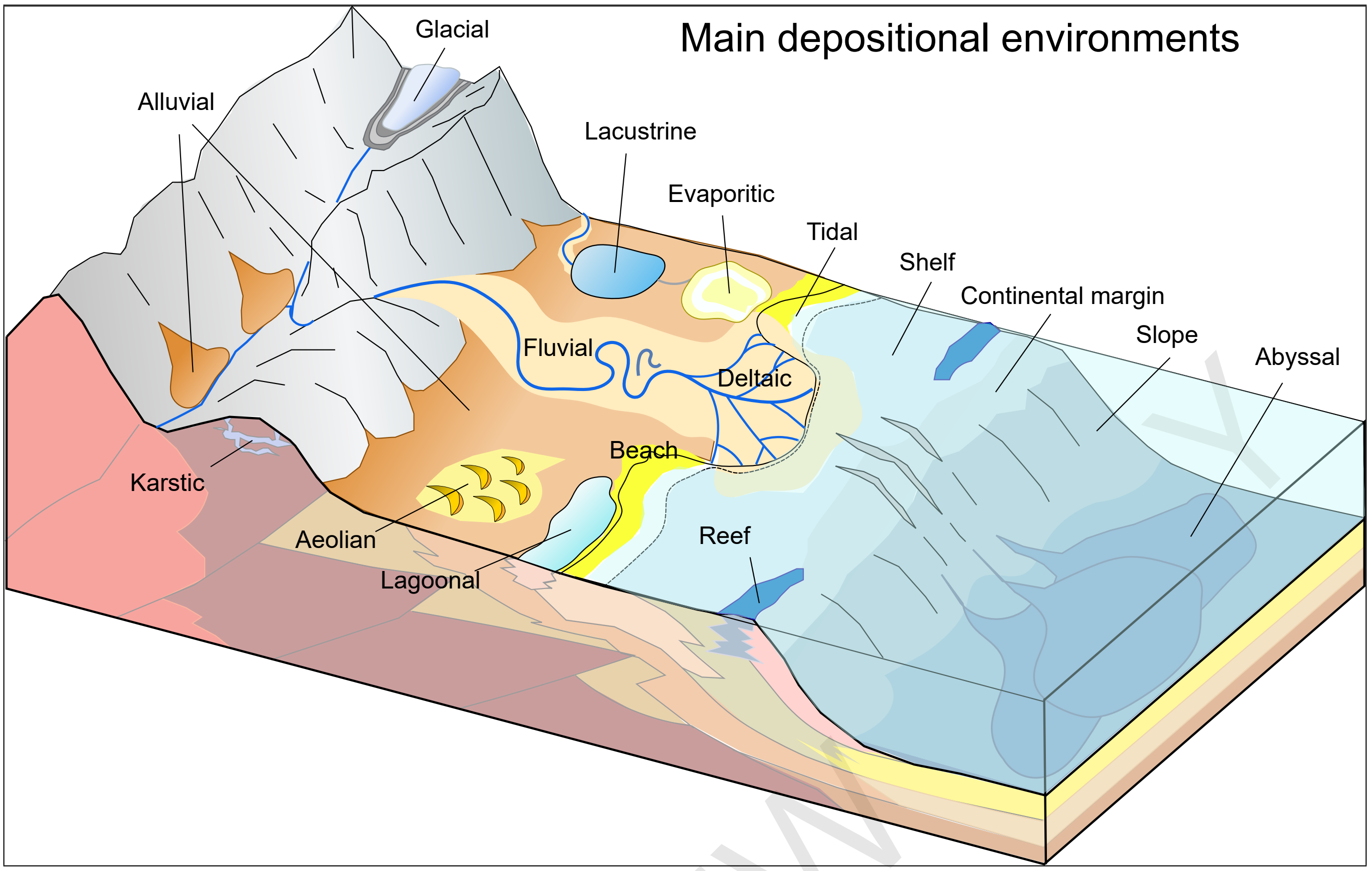

Figure 2 – Schematic diagram of the main depositional environments in sedimentary systems. Source: Mikenorton (2018)

4.2 Figure design

Figures must be designed with accessibility in mind. Do not rely on colour alone, use symbols or patterns to assist with differentiation on graphs and other diagrams.

Preferably use the Noto Sans font family as that is what we use for final production. However, any standard, sans-serif font (e.g. Arial, Helvetica, Verdana) can be used. Be consistent and only use one font family.

Use at most three (preferably no more than two) different font sizes for annotations on figures. The standard font size should be 10 pt Noto Sans or equivalent. Regardless of font choice, the minimum character height must be no less than 1.5 mm (≈ 8 pt Noto Sans) at final production size for accessibility purposes. Aim to use larger font sizes (e.g. 10 pt Noto Sans) to improve accessibility.

Colour schemes **must** be colour-vision deficiency safe (e.g.Scientific Colour Maps).

4.3 Maps

Map figures must have a scale, grid markers of some form, and state the coordinate reference system of their projection. Figure 1 is an exemplar of minimum requirements for map figures.

Due to the nature of geology being an international science, some maps may have areas of disputed borders or regions. In these cases, please use the internationally recognised border or region where possible.

4.4 Indexing and cross-referencing

Figures and tables should be numbered in order of appearance and cross-referenced in text. In LaTeX, use `\autoref{ref}` to handle this. For example, these are cross-references to Figure 2 and Table 1 using `autoref`.

For figures with sub-panels, add uppercase letters for indexing purposes as shown in Figure 1A and Figure 1B. If additional sub-indices are required, use parenthesised lowercase Roman numerals, e.g. A(i).

4.5 Table formatting

Complex tables present a significant challenge in LaTeX for our volunteer production team. Please only use tables where needed, and keep them simple. Complex tables will incur production delays.

Use a table only if there isn't a simpler way to present your content, such as a paragraph of text or figure. Each table must be independently understandable, with a clear title or caption, defined abbreviations, and sufficient explanatory notes where needed. Tables should be numbered sequentially in the order they are cited in the manuscript. Design tables for readability and accessibility. The minimum font size should be Noto Sans 8 pt (or equivalent). Keep tables simple:

- use simple structure, clear column headings, consistent terminology, and SI units;
- avoid excessive decimal places, duplicated information, empty cells without explanation, and unnecessary formatting;
- do not use colour alone to communicate meaning.

Large or highly detailed datasets should be deposited in an appropriate repository and cited in the data availability statement rather than included as oversized manuscript tables. Datasets provided in a form such as CSV, ODS, XLSX, RData, JLD2, HDF5 etc. are better for study reproducibility.

Tables rules are formatted using booktabs. This means you can use the `\toprule`, `\midrule`, and `\bottomrule` commands within the `tabular` environment.

The nicematrix package is also loaded, providing customisation for blocks and the `X` for the flexible column widths.

Table 1 is a simple example of a one-column wide table with correct styling and using the `NiceTabular*` environment.

To ensure correct widths of tables, please use the provided custom commands

- `\onecolwidth` for single-column width
- `\twocolwidth` for two-column width

as the size of your `NiceTabular*` or `tabular*` environment.

For tables that will extend beyond a single page, `longtable` is available. If the `longtable` needs to be two columns wide, or in landscape orientation, please see the `example_longtable.tex` for an example of how to format these. We have placed the longtable at the end of this build to not clutter the primary information.

To place a table that is only a single column wide (i.e. small tables), use the `table` environment.

Otherwise, use the `table*` environment to place a table that will span both page columns.

NOTE: `nicematrix` and `longtable` will not work in the same table.

5 Mathematics

In general follow the Short Math Guide (PDF).

Use `\vec{x}` for vectors and `\mathbf{M}` for matrices, e.g.

```
\vec{y} = \mathbf{X}\vec{\beta} + \epsilon
```

will print:

$$\vec{y} = \mathbf{X}\vec{\beta} + \epsilon \tag{1}$$

In-line equations will look like this: $x^2 + y^2 = z^2$, while basic display equations will look like the following:

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{i=n} x_j = \frac{x_1 + x_2 \ldots + x_n}{n} \tag{2}$$

Multiline equations will look as follows:

$$\begin{multline}\hat{\rho}^2_{OP}\left(\rho^2\right) = 1 - \frac{n-3}{n-\left(k+1\right)-1}\cdot \\ \left(1-\rho^2\right){}_2F_1\left(1,1;\frac{n-\left(k+1\right)-1}{2};1-\rho^2\right) \tag{3}\end{multline}$$

Finally, aligned equations will look like this:

$$
\begin{aligned}
\sum (x_i - \varepsilon_1)(x_i - \varepsilon_2)(x_i - \varepsilon_3)(x_i - \varepsilon_4) &= 0 && (4)\\
\sum x_i (x_i - \varepsilon_1)(x_i - \varepsilon_2)(x_i - \varepsilon_3)(x_i - \varepsilon_4) &= 0 && (5)\\
\sum {x_i}^2 (x_i - \varepsilon_1)(x_i - \varepsilon_2)(x_i - \varepsilon_3)(x_i - \varepsilon_4) &= 0 && (6)\\
\sum {x_i}^3 (x_i - \varepsilon_1)(x_i - \varepsilon_2)(x_i - \varepsilon_3)(x_i - \varepsilon_4) &= 0 && (7)
\end{aligned}
$$

6 Chemical equations

The mhchem package is enabled in the template. Use `\ce{}` to wrap your chemistry. For example, `\ce{^{87}_{37}{Rb}+}` prints $^{87}_{37}Rb^{+}$.

7 Referencing

Appropriate citation of work is mandatory. While peer-reviewed literature should be the primary sources on which you rely, material considered to be *grey literature* such as pre-prints, theses, government reports, conference papers etc. should be appropriately cited.

The *Sedimentologika* class uses the citation-style-langauge package for referencing.

Use `\cite{}` for parenthetical citations, e.g. (Strata, 2065), and `\textcite{}` for in-line text citations, e.g. Helios et al. (2072) stated that this is an awesome template. Prefixes, suffices, and pointers can be added as optional arguments, e.g. `\cite[prefix = {e.g. }, suffix={and references therein}]{ExampleThesis}` will produce (e.g. Diamond, 2060 and references therein).

The following is to enable generation of a reference list with examples for an article (Helios et al., 2072), book (Strata, 2065), book section (Core, 2068), dataset (Cryo & Ice, 2075), map (Carto, 2069), preprint (Plume & Geyser, 2074), report (Agency for Outer Worlds Geology, 2070), software (OrbitLab, 2073) and thesis (Diamond, 2060).

8 Conclusions

This is a mandatory section, replace this text.

Thanks for using the *Sedimentologika* LaTeX template for preparing your article.

Acknowledgements

Replace this text with your acknowledgements.

Data and code availability

Data used in this paper is hosted on Zenodo at doi.org/10.1000/182, `\cite{ExampleCitation}`.

The code used in this paper is available via and archived on Zenodo at doi.org/10.1000/182, `\cite{ExampleCitation}`.

Supplementary materials

If you have any additional supplementary material hosted in a repository for this manuscript please replace this with a statement linking to the doi, and add the citation, e.g. Supplementary material for this paper can be found at doi.org/10.1000/182 `\cite{ExampleCitation}`.

Conflict of interest statement

The authors declare they do not have any tangible or perceived conflicts of interest with regard to this research.

Artificial intelligence use statement

All research, analysis, writing, and editing were performed without generative AI assistance.

Funding

Declare sources of funding for this research.

References

Agency for Outer Worlds Geology. (2070). *Example report: Io volcanic activity annual summary* (EX-2070-01). https://example.com/io-report

Carto, T. (2069). *Example map of Titan hydrocarbon lakes (Scale 1:5,000,000)*. https://example.com/titan-map

Core, P. (2068). Example chapter on Mercury mantle differentiation. In I. Iron & N. Silicate (Eds.), *Example volume: Planetary interiors* (pp. 55–88). Example Academic. https://doi.org/10.1000/182

Cryo, F., & Ice, G. (2075). *Example dataset: Europa ice shell thickness and fracture mapping (Version 1.0)* [Dataset]. Example Data Repository. https://example.com/europa-dataset

Diamond, B. (2060). *Example thesis: Saturnian ring–moon interactions and surface geology* [Doctoral thesis, The University of Saturn]. https://doi.org/10.1000/182

Helios, A., Basalt, C., & Regolith, D. (2072). Example study of volcanic resurfacing on Venusian plains. *Journal of Planetary Geology*, *58*(3), 145–178. https://doi.org/10.1000/182

Mikenorton. (2018). *Schematic diagram showing types of depositional environments* [Graphic]. Wikimedia Commons. https://commons.wikimedia.org/wiki/File:Main_depositional_environments.svg

OrbitLab. (2073). *Example software: NeptuneSeismoSim (Version 1.0)* [Julia Programming Language]. https://example.com/neptune-software

Plume, R., & Geyser, S. (2074). *Example preprint on Enceladus cryovolcanic flux*. https://example.com/enceladus-preprint

Strata, L. (2065). *Example text: Martian sedimentology and stratigraphy*. Example Press. https://doi.org/10.1000/182

Table 2 – Summary of laser conditions and materials analysed in each session.

Session Date	Material	Spot Diameter (μm)	Repetition Rate (Hz)	Nominal Fluence ($J\,cm^{-2}$)	Measured Fluence ($J\,cm^{-2}$)	Ablation Time (s)
2020-02-24	zircon	29	5	2		30
2020-02-24	glass	29	5	3.5		30
2020-02-24	glass	51	5	3.5		30
2020-02-26	zircon	29	5	2		30
2020-02-26	glass	29	5	3.5		30
2020-02-26	glass	51	5	3.5		30
2020-05-06	zircon	19	5	2		40
2020-05-06	glass	19	5	3.5		40
2020-05-06	glass	51	5	3.5		40
2020-05-08	glass	43	5	3.5		40
2020-05-11	zircon	29	5	2		40
2020-05-11	glass	29	5	3.5		40
2020-05-11	glass	51	5	3.5		40
2021-03-30	zircon	29	5	2		30
2021-03-30	glass	43	5	3.5		30
2021-03-31	zircon	30	5	2		40
2021-03-31	glass	43	5	3.5		40
2021-05-06	zircon	30	5	2		40
2021-05-06	glass	43	5	3.5		40
2021-09-06	zircon	30	5	2		30
2021-09-06	glass	43	5	3.5		30
2021-09-06	monazite	20	5	2		30
2022-01-19	baddeleyite	20	5	2		30
2022-01-19	zircon	20	5	2		30
2022-01-19	glass	43	5	3.5		30
2022-01-19	apatite	30	5	3.5		30
2022-02-01	monazite	13	5	2	1.9	30
2022-02-01	xenotime	13	5	2	1.9	30

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Session Date	Material	Spot Diameter (μm)	Repetition Rate (Hz)	Nominal Fluence ($J\,cm^{-2}$)	Measured Fluence ($J\,cm^{-2}$)	Ablation Time (s)
2022-02-01	glass	43	5	3.5	3.6	30
2022-04-01	glass	43	5	3.5	3.4	30
2022-04-21	apatite	30	5	3.5	3.4	30
2022-04-21	zircon	30	5	2	2	30
2022-04-21	glass	43	5	3.5	3.4	30
2022-05-31	glass	43	5	5	5.2	30
2022-05-31	monazite	13	5	2	1.9	30
2022-05-31	xenotime	13	5	2	1.9	30
2022-06-20	zircon	30	5	2	2.1	30
2022-06-20	glass	43	5	3.5	3.6	30
2022-06-20	xenotime	13	5	2	2.1	30
2022-06-29	glass	43	5	3.5	3.4	30
2022-06-29	xenotime	13	5	2	1.9	30
2022-07-08	glass	43	5	3.5	3.6	40
2022-07-08	rutile	43	5	5	5.2	40
2022-08-30	monazite	20	5	2	2	30
2022-08-30	zircon	20	5	2	2	30
2022-10-10	glass	43	5	3.5	3.4	30
2022-10-10	zircon	30	5	2	1.9	30
2022-10-11	glass	30	5	3.5	3.4	30
2022-10-11	zircon	43	5	2	1.9	30
2022-12-09	glass	43	5	3.5	3.5	40
2022-12-09	rutile	43	5	5	4.9	40
2023-02-20	apatite	30	5	3.5	3.5	30
2023-02-20	glass	30	5	3.5	3.5	30
2023-02-20	zircon	30	5	2	1.9	30
2023-03-22	monazite	13	5	2	1.8	30
2023-03-22	xenotime	13	5	2	1.8	30

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Session Date	Material	Spot Diameter (µm)	Repetition Rate (Hz)	Nominal Fluence ($J\,cm^{-2}$)	Measured Fluence ($J\,cm^{-2}$)	Ablation Time (s)
2023-04-21	glass	43	5	3.5	3.5	30
2023-04-21	apatite	43	5	3.5	3.5	30
2023-04-21	zircon	30	5	2	2	30
2023-05-09	glass	43	5	3.5	3.5	40
2023-05-23	glass	43	5	3.5	3.6	40
2023-05-29	glass	43	5	3.5	3.4	30
2023-05-29	monazite	13	5	2	2.1	30
2023-06-30	glass	30	5	3.5	3.4	30
2023-06-30	apatite	30	5	3.5	3.4	30
2023-07-11	glass	43	5	3.5	3.5	40
2023-07-11	zircon	20	5	2	1.9	40
2023-11-24	apatite	30	5	3.5	3.5	30
2023-11-24	baddeleyite	30	5	2	2	30
2023-11-24	monazite	20	5	2	2	30
2023-11-24	rutile	43	5	5	4.9	30
2023-11-24	titanite	43	5	5	4.9	30
2023-11-24	xenotime	20	5	2	2	30
2023-11-24	zircon	30	5	2	2	30
2023-11-24	glass	30	5	3.5	3.5	30
2024-04-29	apatite	30	5	3.5	3.5	40
2024-04-29	baddeleyite	30	5	2	2.1	40
2024-04-29	cassiterite	43	5	5	5.1	40
2024-04-29	monazite	20	5	2	2.1	40
2024-04-29	rutile	43	5	5	5.1	40
2024-04-29	titanite	43	5	5	5.1	40
2024-04-29	xenotime	20	5	2	2.1	40
2024-04-29	zircon	30	5	2	2.1	40
2024-04-29	glass	30	5	3.5	3.5	40